

Multi-Step Prompt Workflow

Theme: Library Management Portal

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Assessment Objective

Design a 3-step prompt workflow in an AI interface for a Library Management Portal. Each step must use the output of the previous step.

Step	Prompting Technique	Purpose
1	Few-shot prompting	Extract library borrowing details
2	CoT reasoning	Determine request priority
3	JSON mode	Create structured borrowing result

Prompting Techniques Used

Step	Prompting Technique	Purpose
1	Few-shot prompting	Extract library borrowing details
2	CoT reasoning	Determine request priority
3	JSON mode	Create structured borrowing result

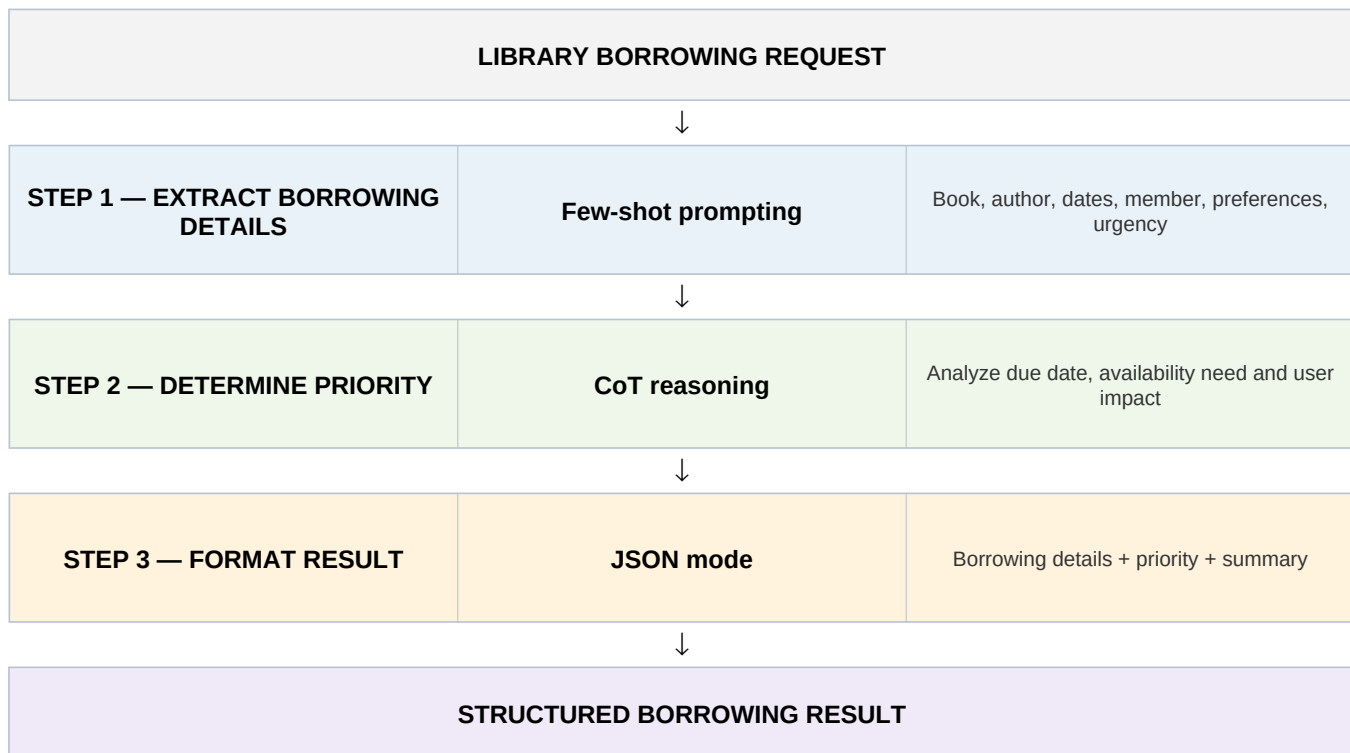
1. Few-shot prompting — examples teach the AI a consistent extraction format.
2. CoT reasoning — the second step evaluates urgency and user impact before assigning a priority.
3. JSON mode — the final step converts the previous outputs into a machine-readable result.

Key requirement: chaining

The output from Step 1 becomes the input to Step 2, and the outputs from Steps 1 and 2 become the inputs to Step 3. No step uses a fixed or hardcoded result.

1. Workflow Overview

The workflow follows the Extract → Reason → Format pattern. A student's natural-language library request is transformed into a structured borrowing decision.



Why this design? Few-shot examples make extraction consistent, reasoning is separated into its own step, and JSON formatting makes the final result easy for a library portal to process.

2. Step 1 — Extract Library Borrowing Details

Technique: Few-shot prompting

The first prompt converts a natural-language library request into consistent borrowing fields.

You are a Library Management Assistant.

Task: Extract the important borrowing details from the student's request.

Return only this bullet format:

- Student: - Book: - Author: - Library/Branch:
- Borrow Date: - Return Date: - Membership Type:
- Preferences: - Special Requirement:

Example 1: Request: "I want to borrow Clean Code by Robert Martin from the main library from 10 September to 20 September. I prefer a digital copy if available."

Output: Student=student | Book=Clean Code | Author=Robert Martin | Branch=main library | Borrow=10 September | Return=20 September | Membership=not stated | Preferences=digital copy | Special Requirement=none stated

Example 2: Request: "Please reserve Database System Concepts for me. I need it urgently because my project review is this week."

Output: Student=student | Book=Database System Concepts | Author=not stated | Branch=not stated | Borrow=not stated | Return=not stated | Membership=not stated | Preferences=none stated | Special Requirement=urgent reservation

Now extract details from: {LIBRARY_BORROWING_REQUEST}

Intermediate output: The extracted borrowing details are passed directly to Step 2.

3. Step 2 — Determine Borrowing Priority

Technique: Chain-of-Thought (CoT) reasoning

This step evaluates the extracted information and assigns a priority of low, medium, or high.

You are a senior library coordinator.

Task: Determine the borrowing/reservation priority from the extracted details below.

Reason step by step about:

1. How soon the student needs the book
2. Whether there is a stated return or project deadline
3. Whether the student may miss an important academic activity
4. Whether the request has a strong time-sensitive requirement

Choose exactly one:

- low = no clear deadline or immediate need
- medium = some time sensitivity but no immediate academic risk
- high = urgent deadline, exam/project need, or immediate requirement

STEP 1 OUTPUT: {STEP_1_OUTPUT}

Return: Priority: | Reason:

Important: Step 2 must use the real Step 1 output. It must not use a fixed or hardcoded priority.

Chaining rule

The priority decision must be based on the actual details extracted in Step 1, so different user requests can produce different priorities.

4. Step 3 — Generate Final JSON

Technique: JSON mode / structured output

The final step combines the extracted borrowing details and priority into a clean machine-readable object.

Return ONLY valid JSON. Do not add Markdown or any explanation outside the JSON.

Required schema:

```
{
  "borrowing_details": {
    "student": "...",
    "book": "...",
    "author": "...",
    "library_branch": "...",
    "borrow_date": "...",
    "return_date": "...",
    "membership_type": "...",
    "preferences": [...],
    "special_requirement": "..."
  },
  "priority": "low|medium|high",
  "summary": "..."
}
```

STEP 1 – EXTRACTED DETAILS: {STEP_1_OUTPUT}

STEP 2 – PRIORITY ASSESSMENT: {STEP_2_OUTPUT}

Rules: Use information from both previous steps. Keep priority exactly low, medium, or high. Preferences must be an array. Output must be valid JSON and directly parseable.

5. End-to-End Test Run

Test Input

Hi, I am Karthik. I need to borrow Clean Code by Robert Martin from the main library from 10 September to 20 September. I have a project review this week, so please arrange the book quickly. I prefer a digital copy if available.

Step 1 — Expected Extraction

- Student: Karthik
- Book: Clean Code
- Author: Robert Martin
- Library/Branch: main library
- Borrow Date: 10 September
- Return Date: 20 September
- Membership Type: not stated
- Preferences: digital copy
- Special Requirement: urgent because project review is this week

Step 2 — Expected Priority

Priority: high

The student has a near-term academic deadline and needs the book quickly for a project review.

Step 3 — Expected Final JSON

```
{
  "borrowing_details": {
    "student": "Karthik",
    "book": "Clean Code",
    "author": "Robert Martin",
    "library_branch": "main library",
    "borrow_date": "10 September",
    "return_date": "20 September",
    "membership_type": "not stated",
    "preferences": ["digital copy"],
    "special_requirement": "Urgent because project review is this week"
  },
  "priority": "high",
  "summary": "Karthik needs Clean Code from the main library for a project review and requests quick access."
}
```

6. Assessment Checklist & Rubric

Criterion	Success condition
Step 1 — Few-shot	Uses examples to produce consistent library borrowing fields.
Step 2 — CoT	Reasoning evaluates the actual Step 1 output to assign priority.
Step 3 — JSON	Final result is valid JSON with borrowing_details, priority and summary.
Chaining	Each step receives the previous step's actual output.
End-to-end test	Workflow is demonstrated using at least one borrowing request.
Transparency	Intermediate outputs are shown for grading.

Conclusion

This Library Management Portal workflow demonstrates prompt chaining through Extract → Reason → Format. It combines few-shot prompting, persona instructions, CoT reasoning, JSON formatting, and intermediate-output chaining in a clear UI-based assessment.